

WERNIGERODE, Germany

WMO: 104540

Lat: 51.8456N

Lon: 10.7686E

Elev: 234

StdP: 98.54

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.6	-10.9	-17.6	0.8	-12.7	-14.5	1.1	-10.0	14.4	6.5	13.2	5.3	1.1	330	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	28.0	18.4	26.3	17.7	24.6	16.9	19.4	26.0	18.5	24.7	17.6	23.4	3.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	12.5	22.3	15.9	11.6	21.2	15.0	11.0	20.6	56.1	25.6	53.1	24.7	50.4	23.5	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.4	10.6	9.1	DB	-16.3	31.8	5.6	1.5	-20.3	32.9	-23.6	33.8	-26.8	34.6	-30.9	35.8	(4)
(5)			WB	-16.2	21.4	5.5	1.6	-20.2	22.6	-23.4	23.5	-26.5	24.4	-30.5	25.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	0.8	-0.5	3.8	7.3	11.9	14.9	17.3	17.2	13.8	9.7	4.6	2.4
	DBStd	7.35	5.54	5.88	4.30	3.92	3.43	3.30	3.15	3.09	2.95	3.75	4.24	4.51
	HDD10.0	1352	284	294	196	100	20	2	0	0	3	51	166	238
	HDD18.3	3596	542	527	452	332	199	112	59	60	139	268	412	494
	CDD10.0	861	0	1	2	18	80	148	226	222	117	41	4	2
	CDD18.3	63	0	0	0	0	1	9	26	24	3	1	0	0
	CDH23.3	641	0	0	0	1	21	97	261	227	32	3	0	0
	CDH26.7	121	0	0	0	0	1	17	51	51	1	0	0	0
Wind	WSAvg	3.4	4.4	3.5	3.8	3.3	2.8	2.9	3.0	2.8	3.2	3.3	3.6	4.1
Precipitation	PrecAvg	587	42	34	44	52	65	69	52	61	32	41	50	46
	PrecMax	751	135	87	96	135	124	152	122	100	64	97	91	122
	PrecMin	426	6	6	11	16	33	15	21	30	7	7	22	3
	PrecStd	118	29	23	28	30	27	35	25	23	15	26	24	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.9	14.6	17.2	21.6	25.5	29.0	30.1	31.0	26.4	23.4	16.1	13.7
		MCWB	8.9	8.8	9.9	12.7	15.9	19.1	19.3	19.3	16.9	15.8	11.4	8.5
	2%	DB	9.8	11.5	13.7	18.6	23.4	26.1	28.0	27.8	24.0	19.8	13.6	11.9
		MCWB	7.1	7.5	8.8	11.5	14.8	18.1	18.7	18.0	16.3	14.1	9.9	8.4
	5%	DB	8.6	8.9	11.5	16.2	21.2	23.9	26.3	25.8	21.6	17.3	11.9	10.5
		MCWB	6.2	6.0	7.8	10.0	13.7	16.5	18.0	17.6	15.5	12.9	9.1	7.6
	10%	DB	7.3	7.0	9.7	14.0	19.2	21.7	24.4	23.9	19.3	15.4	10.5	8.9
		MCWB	4.9	4.6	6.7	9.1	13.1	15.8	17.2	16.6	14.3	11.6	8.0	6.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.0	9.6	10.7	13.7	17.5	20.9	20.9	20.2	18.2	16.5	11.8	9.8
		MCDB	11.4	14.1	15.1	20.0	22.9	26.2	27.6	28.4	24.3	22.3	13.9	12.8
	2%	WB	7.7	7.8	9.4	12.0	15.8	18.7	19.6	18.9	17.0	14.6	10.3	8.7
		MCDB	9.7	11.0	13.5	17.7	21.1	24.4	26.1	26.0	22.5	18.8	12.9	11.4
	5%	WB	6.3	6.2	8.1	10.5	14.6	17.4	18.7	18.0	15.8	13.2	9.3	7.7
		MCDB	8.3	8.8	11.0	15.2	20.0	22.6	24.8	24.4	20.6	16.8	11.7	10.2
	10%	WB	5.1	4.7	6.8	9.3	13.4	16.1	17.7	17.1	14.8	12.0	8.2	6.5
		MCDB	7.2	6.9	9.6	13.7	18.7	21.1	23.6	22.7	19.3	15.1	10.5	8.8
Mean Daily Temperature Range	5% DB	MDBR	5.7	6.4	6.8	8.3	9.7	8.9	9.6	9.7	8.4	7.5	5.9	5.2
		MCDBR	6.7	8.4	10.1	11.5	13.8	13.1	13.5	13.2	12.6	11.1	8.3	6.3
		MCWBR	5.6	6.4	6.9	6.8	7.6	7.0	6.7	6.4	7.0	7.3	6.1	5.0
	5% WB	MCDBR	5.9	8.5	9.2	10.3	11.9	11.4	12.3	11.8	10.9	10.1	7.6	5.7
		MCWBR	5.4	6.7	6.6	6.6	7.3	6.6	6.6	6.2	6.5	7.0	5.7	4.9
Clear-Sky Solar Irradiance	taub	0.311	0.334	0.373	0.411	0.414	0.410	0.427	0.420	0.382	0.354	0.332	0.305	
	taud	2.369	2.340	2.251	2.196	2.222	2.256	2.215	2.253	2.347	2.410	2.400	2.368	
	Ebn at Noon	694	779	811	822	839	845	821	806	796	747	660	633	
	Edh at Noon	61	83	109	130	133	129	133	122	99	77	60	52	
All-Sky Solar Radiation	RadAvg	0.70	1.40	2.47	4.00	4.91	5.30	4.97	4.24	3.08	1.77	0.81	0.53	
	RadStd	0.08	0.19	0.28	0.48	0.57	0.43	0.67	0.39	0.38	0.19	0.10	0.06	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page